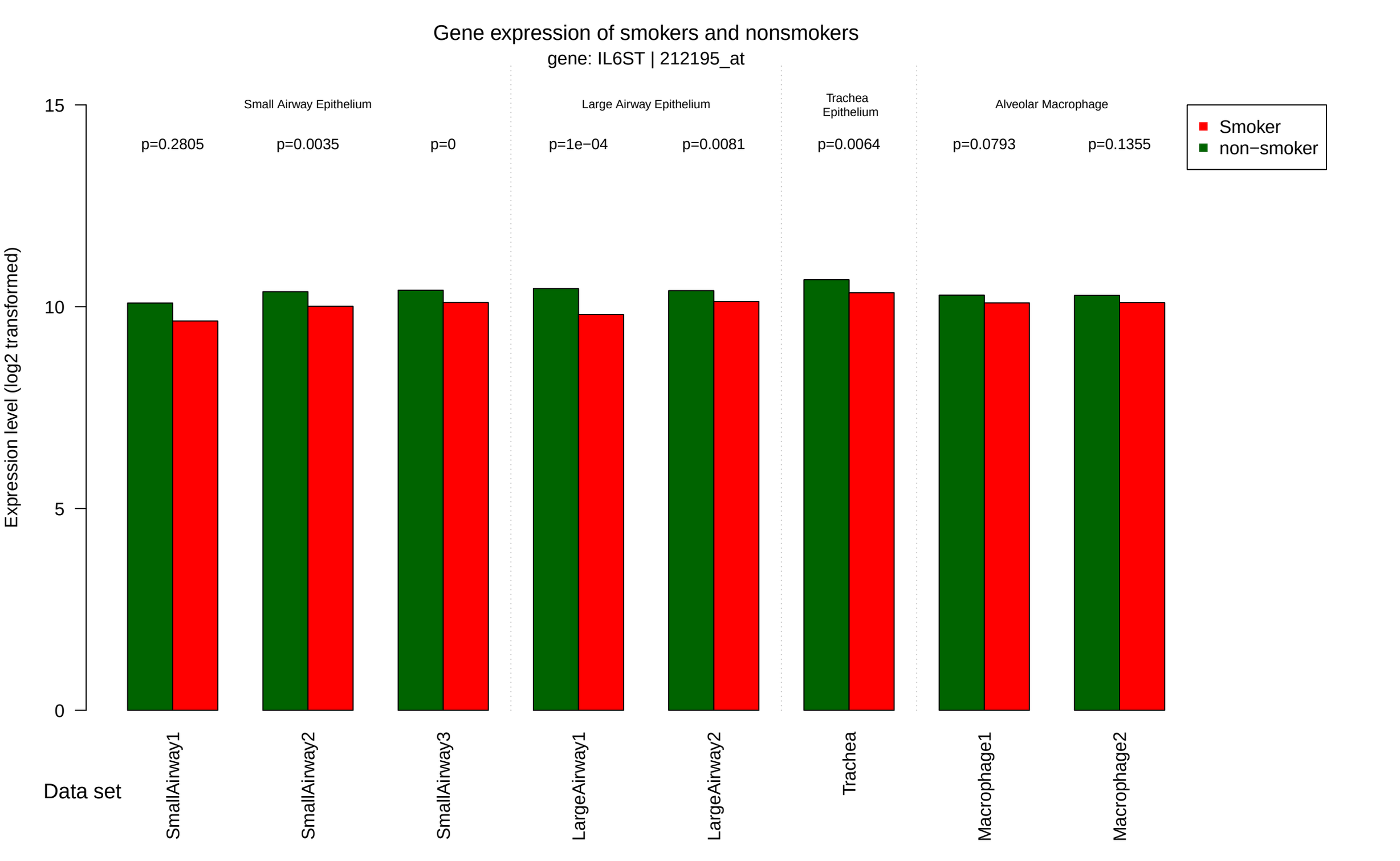
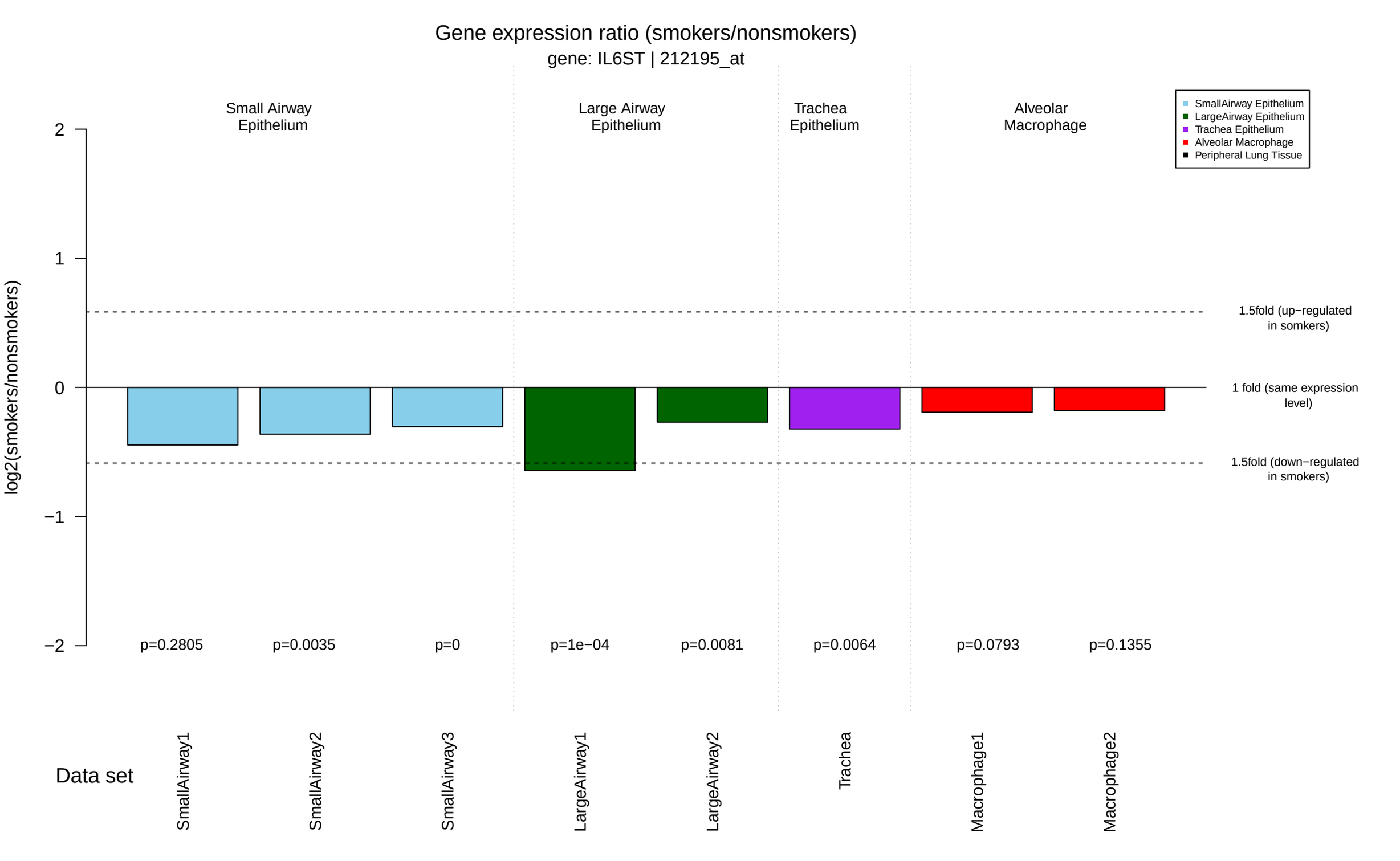
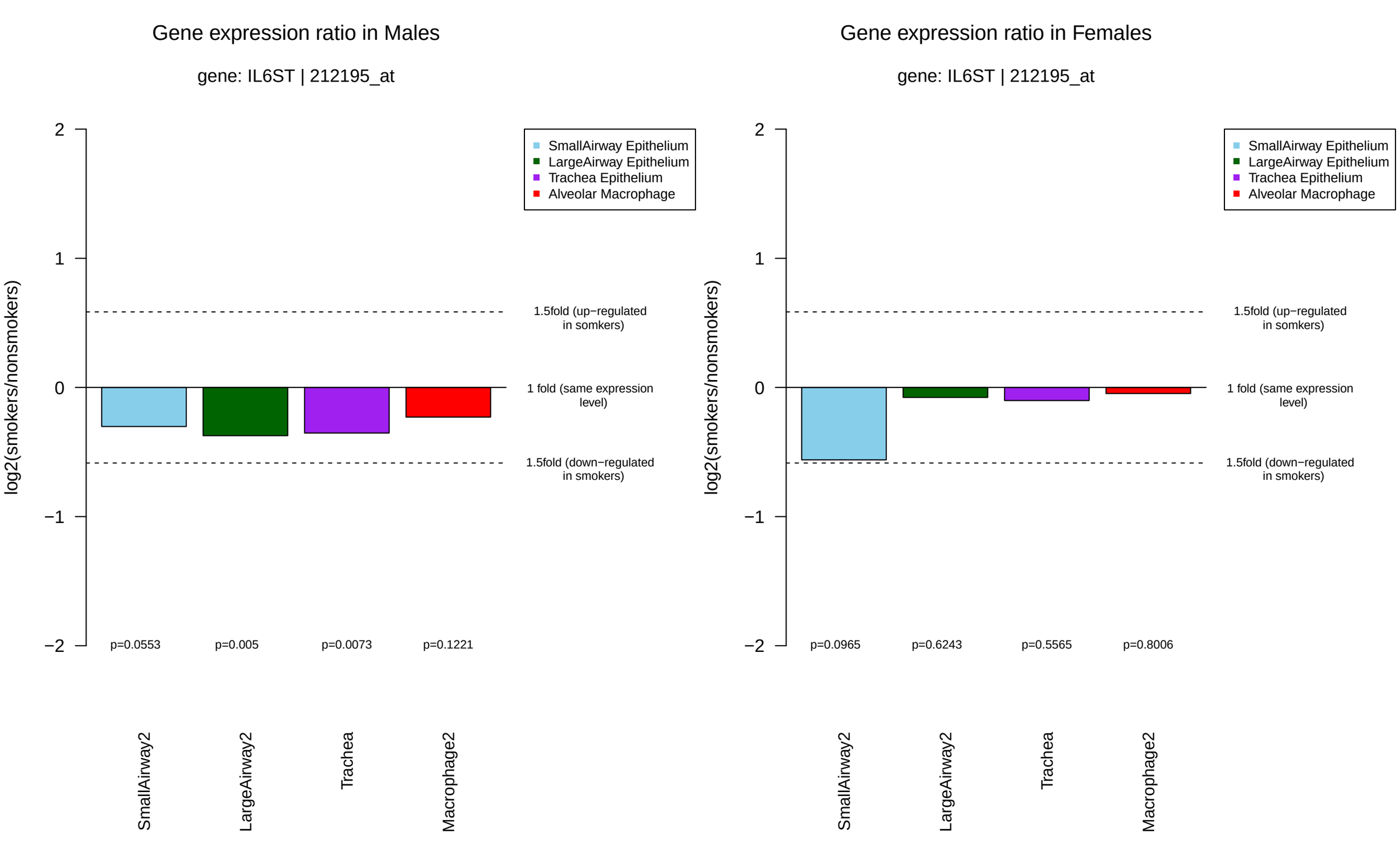
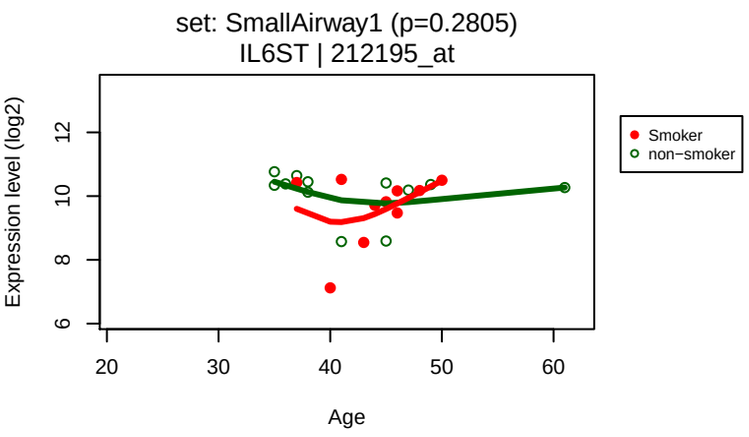
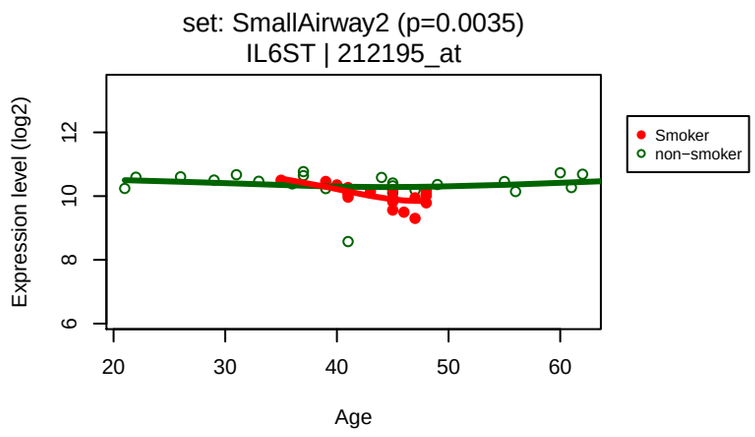
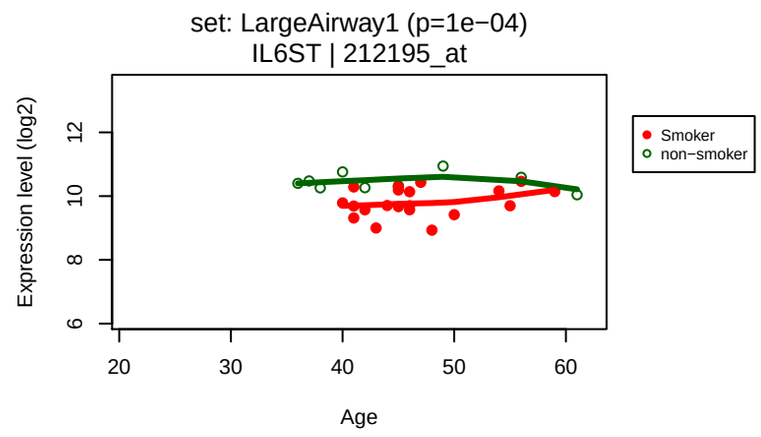

Gene expression of smokers and nonsmokers
gene: IL6ST | 212195_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2805
p=0.0035
p=0
p=1e−04
p=0.0081
p=0.0064
p=0.0793
p=0.1355
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

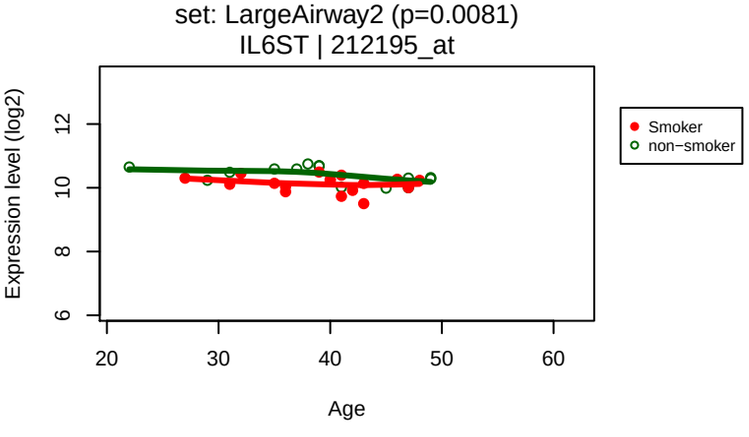

Gene expression ratio (smokers/nonsmokers)
gene: IL6ST | 212195_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2805
p=0.0035
p=0
p=1e−04
p=0.0081
p=0.0064
p=0.0793
p=0.1355
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

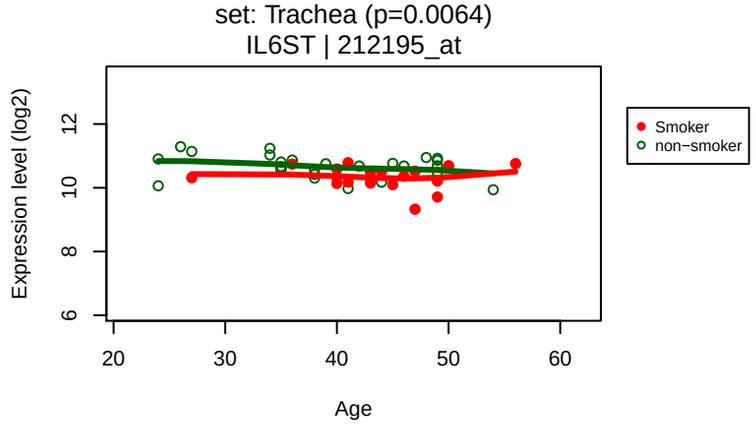

Gene expression ratio in Males
gene: IL6ST | 212195_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0553
p=0.005
p=0.0073
p=0.1221
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IL6ST | 212195_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0965
p=0.6243
p=0.5565
p=0.8006
SmallAirway2
LargeAirway2
Trachea
Macrophage2

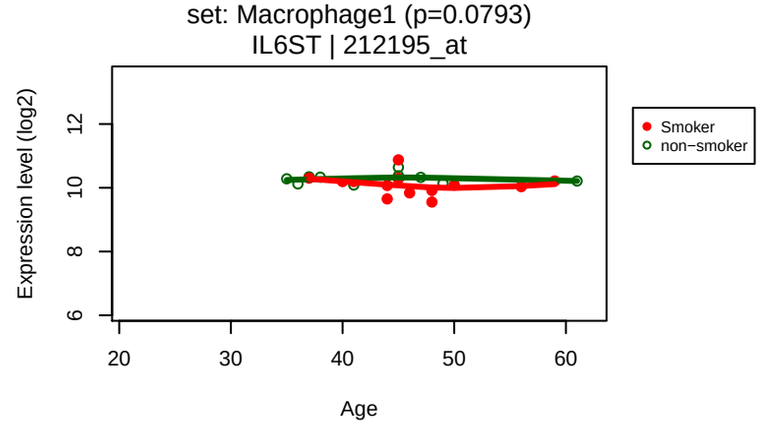

set: SmallAirway1 (p=0.2805)
IL6ST | 212195_at
Expression level (log2)
Age
Smoker
non−smoker

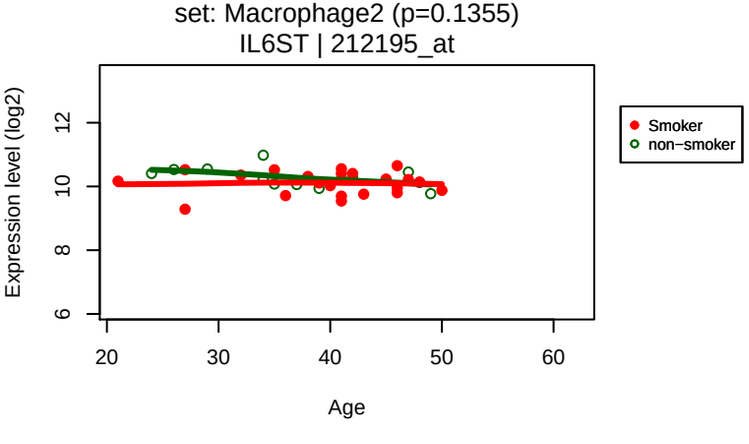

set: SmallAirway2 (p=0.0035)
IL6ST | 212195_at
Expression level (log2)
Age
Smoker
non−smoker

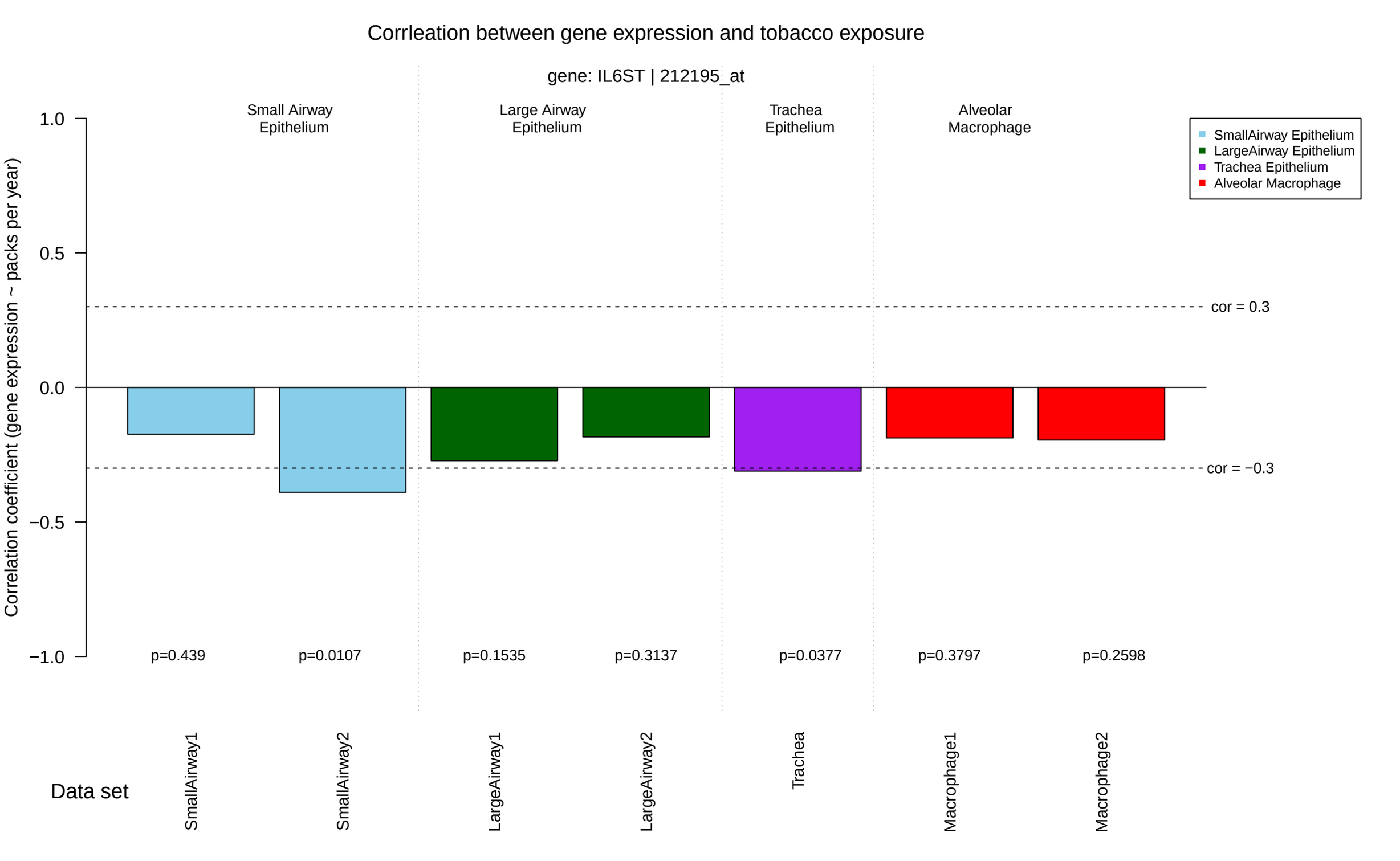

set: LargeAirway1 (p=1e−04)
IL6ST | 212195_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0081)
IL6ST | 212195_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0064)
IL6ST | 212195_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0793)
IL6ST | 212195_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1355)
IL6ST | 212195_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IL6ST | 212195_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.439
p=0.0107
p=0.1535
p=0.3137
p=0.0377
p=0.3797
p=0.2598
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2